

EA-0011 - AQELYN Engineering Archive

AQELYN - EA-0011 Engineering Archive

IS-011 - AQELYN Identity & Access Governance Engine

Archive ID: EA-0011

Implementation Specification: IS-011

Component: AQELYN Identity & Access Governance Engine

Project: AQELYN

System Type: Cyber Security Operating Environment

Status: COMPLETE

Repository Impact: No top-level repository structure changes

Breaking Changes: None

Engineering Phase: Phase 3

Predecessor Archives: EA-0001 through EA-0010

Next Specification: IS-012

Document Control

| Field | Value |

|---|

| Document | Engineering Archive EA-0011 |

| Specification | IS-011 - AQELYN Identity & Access Governance Engine |

| Publication Format | Markdown, PDF, HTML, ZIP |

| Source of Truth | MD/EA-0011.md |

| Archive Rule | Implementation Specification -> Engineering Archive -> Continue |

| Repository Rule | Fixed repository structure; no redesign |

| Completion State | IS-011 complete; EA-0011 generated |

1. Engineering Context

AQELYN is being built as a modular Cyber Security Operating Environment.

Completed engineering archive chain:

| Engineering Archive | Implementation Specification | Component |

|---|

| EA-0001 | IS-001 | AQELYN Kernel |

| EA-0002 | IS-002 | Universal Object Model |

| EA-0003 | IS-003 | AQELYN Event Bus |

EA-0004	IS-004	AQELYN Evidence Engine
EA-0005	IS-005	AQELYN Knowledge Graph
EA-0006	IS-006	AQELYN Trust Engine
EA-0007	IS-007	AQELYN Mission Engine
EA-0008	IS-008	AQELYN Workflow Engine
EA-0009	IS-009	AQELYN Policy Engine
EA-0010	IS-010	AQELYN Compliance & Governance Engine
EA-0011	IS-011	AQELYN Identity & Access Governance Engine

The fixed repository structure remains:

```
AQELYN/  
■■■ archive/  
■■■ blueprint/  
■■■ docs/  
■■■ src/  
■■■ tests/  
■■■ tools/  
■■■ build/  
■■■ releases/  
■■■ scripts/  
■■■ assets/  
■■■ examples/  
■■■ plugins/  
■■■ sdk/  
■■■ api/  
■■■ README.md
```

The engineering rule remains:

```
Finish Implementation Specification  
↓  
Generate Engineering Archive  
↓  
Continue
```

No Engineering Archive may be skipped.

2. IS-011 Specification Identity

```
Specification ID: IS-011  
Name: AQELYN Identity & Access Governance Engine  
Engineering Archive Target: EA-0011  
Project: AQELYN  
System Type: Cyber Security Operating Environment  
Status: Complete  
Predecessor: IS-010 - AQELYN Compliance & Governance Engine
```

3. Purpose

The AQELYN Identity & Access Governance Engine manages identity, access, entitlement, role, privilege, and approval governance across AQELYN.

It answers:

```
Who has access?  
To what?  
Why?  
Approved by whom?  
Based on which policy?  
Supported by what evidence?  
Reviewed when?  
Should access continue?
```

4. Mission

The engine shall provide:

- Identity inventory
- Access governance
- Entitlement mapping
- Role governance
- Privileged access governance
- Access review workflows
- Access certification
- Policy-based access validation
- Evidence-backed access decisions
- Integration with compliance, trust, workflow, policy, and evidence engines

5. Scope

5.1 In Scope

- Identity records
- Account records
- Role definitions
- Entitlement catalog
- Access grants
- Access requests
- Access reviews
- Access certification
- Privileged access governance
- Orphaned account detection
- Stale access detection
- Segregation-of-duties checks
- Policy-to-access mapping
- Evidence-backed approval chains

5.2 Out of Scope

- Password manager implementation
- Identity provider replacement
- Biometric authentication engine
- Full IAM product replacement
- Human HR system replacement
- Legal employment decision engine

6. Dependencies

IS-011 depends on:

- IS-001 AQELYN Kernel
- IS-002 Universal Object Model
- IS-003 AQELYN Event Bus
- IS-004 AQELYN Evidence Engine
- IS-005 AQELYN Knowledge Graph
- IS-006 AQELYN Trust Engine
- IS-007 AQELYN Mission Engine
- IS-008 AQELYN Workflow Engine
- IS-009 AQELYN Policy Engine
- IS-010 AQELYN Compliance & Governance Engine

7. High-Level Architecture

AQELYN Identity & Access Governance Engine

-
- Identity Registry
- Account Registry
- Role & Entitlement Catalog
- Access Grant Manager
- Access Request Manager
- Access Review Service
- Certification Engine
- Privileged Access Governance Service
- Segregation-of-Duties Analyzer
- Access Evidence Binder
- Access Risk Scoring Service
- Governance Decision Connector
- Identity Event Publisher

8. Functional Requirements

FR-011-001 - Identity Registry

The system shall maintain governed identity records for:

```
Human users
Service accounts
Machine identities
API clients
Workload identities
Temporary identities
External identities
```

FR-011-002 - Account Registry

The system shall map identities to accounts across systems.

Each account shall include:

```
account_id
identity_id
system_id
account_type
status
created_at
last_seen
owner
authentication_method
linked_roles
linked_entitlements
risk_score
```

FR-011-003 - Role & Entitlement Catalog

The system shall maintain a catalog of roles and entitlements.

Each entitlement shall include:

```
entitlement_id
system_id
name
description
permission_scope
sensitivity
risk_level
owner
approval_required
review_frequency
```

FR-011-004 - Access Grant Management

The system shall track access grants between identities, accounts, roles, entitlements, assets, and systems.

FR-011-005 - Access Request Workflow

The system shall support access request workflows including:

```
request
justification
policy evaluation
risk evaluation
approval
provisioning signal
evidence capture
expiration
review
```

FR-011-006 - Access Reviews

The system shall support periodic access reviews.

Review outcomes:

```
keep
remove
modify
escalate
exception
unknown
```

FR-011-007 - Access Certification

The system shall support certification campaigns for:

```
Privileged users
Sensitive systems
High-risk entitlements
Compliance controls
Mission-critical assets
External users
Service accounts
```

FR-011-008 - Privileged Access Governance

The system shall identify, monitor, and govern privileged access.

FR-011-009 - Orphaned and Stale Access Detection

The system shall detect:

```
orphaned accounts
inactive accounts
stale entitlements
unowned service accounts
expired temporary access
access without approval evidence
```

FR-011-010 - Segregation of Duties

The system shall identify access combinations that violate segregation-of-duties rules.

9. Non-Functional Requirements

Traceability
 Immutability of approval records
 Least privilege
 Separation of duties
 Auditability
 Policy explainability
 Evidence-backed access decisions
 Scalable identity graph processing
 Event-driven synchronization
 Backward compatibility

10. Core Governance Flow

```

Identity discovered
↓
Accounts linked
↓
Roles and entitlements mapped
↓
Access evaluated against policy
↓
Risk and trust scored
↓
Approval or remediation workflow triggered
↓
Evidence captured
↓
Access certified
↓
Governance event published
  
```

11. Internal Component Architecture

The Identity & Access Governance Engine is composed of independent services communicating through the AQELYN Event Bus.

```

AQELYN Identity & Access Governance Engine
■
■■■ Identity Registry
■■■ Account Registry
■■■ Identity Correlation Service
■■■ Role & Entitlement Catalog
■■■ Access Grant Manager
■■■ Access Request Manager
■■■ Access Review Engine
■■■ Certification Manager
■■■ Privileged Access Governance Service
■■■ Segregation-of-Duties Analyzer
■■■ Access Risk Engine
■■■ Identity Evidence Service
■■■ Governance Connector
■■■ Reporting Service
■■■ Event Publisher
  
```

Each service shall be independently deployable and testable.

12. Component Specifications

12.1 Identity Registry

Maintains authoritative identity objects.

Supported identity types:

- Employee
- Contractor
- Partner
- Customer
- Service Account
- Application
- Machine Identity
- API Client
- Temporary Identity
- External Identity

Responsibilities:

- Identity lifecycle
- Identity ownership
- Identity metadata
- Identity status
- Identity relationships

12.2 Account Registry

Stores accounts belonging to identities.

Supported systems include:

- Active Directory
- Azure AD
- LDAP
- Linux
- Windows
- Databases
- Cloud Platforms
- SaaS Applications
- Containers
- Kubernetes
- Custom Systems

Each account shall be linked to exactly one governing identity unless explicitly classified as shared or anonymous.

12.3 Identity Correlation Service

Responsible for identity resolution.

Example:

```
graph TD
    A[John Smith] --> B[Azure AD]
    B --> C[Active Directory]
    C --> D[GitHub]
    D --> E[VPN]
    E --> F[Linux]
    F --> G[Single Identity Object]
```

The correlation engine shall support deterministic and configurable matching rules.

12.4 Role & Entitlement Catalog

Maintains normalized definitions for:

- Business Roles
- Technical Roles
- Application Roles
- Administrative Roles

Entitlements
Permissions
Privileges
Groups

12.5 Access Grant Manager

Maintains relationships:

Identity
↓
Account
↓
Role
↓
Entitlement
↓
Permission
↓
Asset

Supports:

Grant
Modify
Suspend
Revoke
Expire
Delegate

12.6 Access Request Manager

Responsible for lifecycle management of access requests.

Lifecycle:

Request
↓
Policy Validation
↓
Risk Evaluation
↓
Approval Workflow
↓
Provisioning Signal
↓
Evidence Collection
↓
Review
↓
Expiration

12.7 Access Review Engine

Supports periodic certification.

Review sources:

Manager Review
Application Owner
System Owner
Compliance Review
Mission Owner
Governance Board

Possible outcomes:

Approved
Revoked
Modified

Deferred
Escalated

12.8 Certification Manager

Supports campaign-based certification.

Campaign types:

Quarterly
Annual
Privileged Users
Mission Critical
Compliance Driven
Emergency Review

12.9 Privileged Access Governance Service

Maintains inventory of privileged identities.

Examples:

Domain Administrators
Cloud Administrators
Database Administrators
Root Accounts
Break Glass Accounts
Emergency Accounts

Capabilities:

Inventory
Approval
Monitoring
Certification
Risk Analysis
Evidence Collection

12.10 Segregation-of-Duties Analyzer

Detects conflicting privileges.

Example:

Create Vendor
+
Approve Vendor Payment
↓
Conflict

The analyzer shall support configurable rule sets.

12.11 Access Risk Engine

Calculates access risk using:

Identity Risk
Trust Score
Entitlement Sensitivity
Mission Criticality
Compliance Requirements
Previous Findings
Behavior Indicators

Outputs:

Low
Moderate
High
Critical

12.12 Identity Evidence Service

Links access decisions to evidence.

Sources include:

Approval Records
Policy Decisions
Authentication Events
Provisioning Records
Review Decisions
Certification Results
Workflow History

13. Universal Object Model Extensions

The following objects extend IS-002.

13.1 Identity

Identity:
identity_id
identity_type
display_name
owner
status
trust_score
risk_score

13.2 Account

Account:
account_id
identity_id
platform
username
status
last_seen

13.3 Role

Role:
role_id
role_type
owner
description

13.4 Entitlement

Entitlement:
entitlement_id
system
privilege
sensitivity

13.5 AccessGrant

AccessGrant:
grant_id
identity
account

```
role
entitlement
approval
```

13.6 AccessReview

```
Review:
review_id
reviewer
decision
timestamp
```

13.7 CertificationCampaign

```
Campaign:
campaign_id
type
owner
due_date
status
```

14. Knowledge Graph Integration

Relationships include:

```
Identity
↓
owns
↓
Account
↓
receives
↓
Role
↓
contains
↓
Entitlement
↓
protects
↓
Asset
```

Additional relationships:

```
Identity
↓
approved_by
↓
Governance Decision
↓
supported_by
↓
Evidence
```

15. Event Bus Integration

The engine publishes standardized events.

15.1 Identity Events

```
identity.created
identity.updated
identity.disabled
identity.deleted
```

15.2 Account Events

```
account.created  
account.linked  
account.disabled  
account.deleted
```

15.3 Access Events

```
access.requested  
access.granted  
access.revoked  
access.expired  
access.review.completed
```

15.4 Certification Events

```
certification.started  
certification.completed  
certification.failed
```

15.5 Governance Events

```
governance.access.approved  
governance.access.rejected  
governance.sod.violation.detected
```

16. Evidence Engine Integration

Evidence references include:

```
Approval Evidence  
Authentication Logs  
Provisioning Records  
Workflow Logs  
Policy Decisions  
Review Records
```

The Identity Engine shall only reference immutable evidence identifiers.

17. Policy Engine Integration

The Policy Engine determines:

```
Who may request access  
Who may approve access  
Required approval chain  
Required evidence  
Required review frequency  
Expiration rules
```

18. Compliance & Governance Integration

IS-010 consumes identity governance information for:

```
Access Control Compliance  
Privileged Access Reviews  
Segregation of Duties  
Certification Status  
Audit Readiness  
Evidence Completeness
```

19. Workflow Engine Integration

Example workflow:

```
Access Request
↓
Policy Evaluation
↓
Risk Assessment
↓
Approval
↓
Provisioning
↓
Evidence Recording
↓
Periodic Review
↓
Certification
```

20. Trust Engine Integration

Access confidence is calculated using:

```
Identity Trust
Device Trust
Evidence Trust
Behavior Trust
Approval Trust
```

Produces:

```
Access Confidence Score
```

21. Mission Engine Integration

Mission policies may define:

```
Required Roles
Required Certifications
Required Clearance
Required Approval Authority
```

Mission readiness shall consider identity governance where applicable.

22. Public APIs

22.1 Identity API

```
GET /identities
POST /identities
GET /identities/{id}
PUT /identities/{id}
```

22.2 Account API

```
GET /accounts
POST /accounts
```

```
PUT /accounts/{id}
```

22.3 Access API

```
POST /access/request
POST /access/grant
POST /access/revoke
GET /access/history
```

22.4 Certification API

```
GET /certifications
POST /certifications/start
POST /certifications/complete
```

22.5 Governance API

```
GET /governance/access
GET /governance/reviews
```

23. Repository Impact

No repository redesign.

Expected implementation:

```
AQELYN/
■■■ src/
■ ■■■ identity_governance/
■■■ tests/
■ ■■■ identity_governance/
■■■ docs/
■ ■■■ identity_governance/
■■■ api/
■ ■■■ identity_governance/
■■■ archive/
```

This complies with the permanent AQELYN repository standard.

24. Security Architecture

The AQELYN Identity & Access Governance Engine is a Tier-1 security subsystem responsible for governing identities, access rights, entitlements, and privileged accounts across the AQELYN Cyber Security Operating Environment.

All access governance decisions shall be:

```
Policy-driven
Evidence-backed
Fully auditable
Cryptographically traceable where configured
```

24.1 Security Principles

The engine shall implement:

```
Zero Trust
Least Privilege
Need-to-Know
Separation of Duties
Identity Assurance
Continuous Verification
```

Immutable Audit Trail
Evidence-Based Decisions
Defense in Depth

24.2 Authorization Model

Identity governance operations require authenticated and authorized actors.

Supported governance roles include:

- | Role | Responsibilities |
- |--|--|
- | Identity Administrator | Identity lifecycle management |
- | Application Owner | Approve application access |
- | System Owner | Approve system-level access |
- | Compliance Officer | Review certifications |
- | Governance Board | Approve exceptions and high-risk access |
- | Security Administrator | Review privileged identities |
- | Automation Service | Execute approved workflows |

All authorization decisions shall be evaluated through the AQELYN Policy Engine.

24.3 Least Privilege Enforcement

The engine shall continuously evaluate whether granted permissions exceed operational requirements.

Capabilities include:

Privilege minimization
Unused entitlement detection
Excessive permission detection
Privilege expiration
Automatic review triggers

24.4 Identity Integrity

Identity objects shall maintain:

Unique Identifier
Lifecycle State
Ownership
Evidence Links
Trust Score
Risk Score
Version History

Identity records are append-only with respect to governance history.

25. Identity Lifecycle

Every identity follows a governed lifecycle.

Discovered
↓
Registered
↓
Validated
↓
Provisioned

↓
Operational
↓
Reviewed
↓
Certified
↓
Modified
↓
Suspended
↓
Deprovisioned
↓
Archived

25.1 Account Lifecycle

Created
↓
Linked
↓
Provisioned
↓
Active
↓
Reviewed
↓
Disabled
↓
Deleted

25.2 Access Grant Lifecycle

Requested
↓
Policy Evaluation
↓
Risk Assessment
↓
Approval
↓
Provisioned
↓
Evidence Recorded
↓
Periodic Review
↓
Revoked

25.3 Certification Lifecycle

Campaign Created
↓
Reviewer Assignment
↓
Review
↓
Decision
↓
Evidence Collection
↓
Closure

26. Risk Evaluation

Access decisions shall consider:

Identity Risk
Privilege Level
Asset Sensitivity
Mission Criticality

Trust Score
Compliance Requirements
Behavioral Indicators
Previous Findings

Risk classifications:

Low
Moderate
High
Critical

High and Critical risk grants may require additional approval stages as defined by policy.

27. Continuous Governance

The engine shall continuously evaluate:

Dormant accounts
Privileged accounts
Expired access
Temporary access
Shared accounts
Orphaned accounts
Excessive entitlements
Segregation-of-duties conflicts

Detected issues shall generate governance events and, where configured, remediation workflows.

28. Audit & Reporting

The engine shall support generation of:

28.1 Executive Reports

Identity Inventory
Privileged Access Summary
Access Risk Overview
Certification Status

28.2 Operational Reports

Pending Reviews
Expired Access
Inactive Accounts
Provisioning Activity

28.3 Compliance Reports

Certification Coverage
Segregation-of-Duties Violations
Privileged Access Reviews
Evidence Completeness

28.4 Engineering Reports

Identity Relationships
Role Mapping
Entitlement Graph
Workflow Statistics

29. Failure Handling

29.1 Missing Identity

Status:
Unknown Identity

Action:
Reject governance decision

29.2 Missing Evidence

Status:
Incomplete

Action:
Require manual review

29.3 Policy Evaluation Failure

Status:
Pending

Action:
Queue for re-evaluation

29.4 Workflow Failure

Status:
Escalated

Action:
Notify governance authority

29.5 Event Bus Interruption

Identity governance events shall be queued until reliable delivery is restored.

30. Performance Requirements

The engine shall support:

Incremental identity synchronization
Event-driven updates
Cached entitlement metadata
Parallel identity correlation
Asynchronous certification campaigns
High-volume review processing

Identity synchronization shall not block governance workflows.

31. Scalability Requirements

The architecture shall support:

Millions of identities
Millions of accounts
Millions of access grants
Large entitlement catalogs
Large certification campaigns
Multi-tenant deployments

Distributed governance processing

32. Testing Strategy

32.1 Unit Testing

Validate:

Identity registry
Account registry
Correlation engine
Risk engine
Access reviews
Certification engine
SoD analyzer

32.2 Integration Testing

Verify interaction with:

AQELYN Kernel
Event Bus
Evidence Engine
Knowledge Graph
Trust Engine
Workflow Engine
Policy Engine
Compliance Engine
Mission Engine

32.3 System Testing

Validate complete scenarios including:

Identity discovery
Account correlation
Access request
Approval workflow
Provisioning signal
Certification campaign
Privileged access review
Governance reporting

32.4 Security Testing

Verify:

Authorization
Least privilege enforcement
Separation of duties
Identity integrity
Audit trail completeness
Privileged access protection

32.5 Regression Testing

Ensure IS-001 through IS-010 continue to operate without behavioral changes introduced by IS-011.

33. Acceptance Criteria

IS-011 shall be considered complete when:

```
Identity registry is defined.
Account registry is defined.
Role and entitlement catalog is defined.
Access governance workflows are documented.
Certification lifecycle is implemented.
Privileged access governance is supported.
Segregation-of-duties analysis is defined.
Risk evaluation model is documented.
Integration with IS-001 through IS-010 is defined.
Repository structure remains unchanged.
Testing strategy is documented.
```

34. Repository Validation

This specification introduces no changes to the agreed repository layout.

Implementation is expected under:

```
AQELYN/
■■■ src/identity_governance/
■■■ tests/identity_governance/
■■■ docs/identity_governance/
■■■ api/identity_governance/
■■■ archive/
```

The top-level repository remains fixed.

35. Completion Summary

IS-011 - AQELYN Identity & Access Governance Engine introduces a comprehensive identity governance subsystem that:

```
Governs identities, accounts, roles, and entitlements.
Supports access request, approval, certification, and review workflows.
Detects segregation-of-duties conflicts.
Governs privileged access.
Integrates with compliance, policy, trust, evidence, workflow, mission, and knowledge graph engines.
Preserves repository stability and backward compatibility.
```

36. Specification Status

```
Specification ID : IS-011
Title : AQELYN Identity & Access Governance Engine
Status : COMPLETE
Engineering Archive : READY FOR GENERATION
Next Artifact : EA-0011
```

Engineering workflow status:

```
EA-0001 COMPLETE
EA-0002 COMPLETE
EA-0003 COMPLETE
EA-0004 COMPLETE
EA-0005 COMPLETE
EA-0006 COMPLETE
EA-0007 COMPLETE
EA-0008 COMPLETE
EA-0009 COMPLETE
EA-0010 COMPLETE
IS-011 COMPLETE
EA-0011 READY TO GENERATE
```

37. EA-0011 Engineering Objective

The objective of IS-011 was to introduce a dedicated Identity & Access Governance Engine that enables AQELYN to govern identities, accounts, roles, entitlements, access grants, access requests, access reviews, privileged access, certifications, and segregation-of-duties controls.

The engine extends AQELYN governance from compliance status into practical access accountability and identity risk management.

38. EA-0011 Engineering Summary

The implementation specification defines a modular subsystem responsible for:

- Identity registry management
- Account registry management
- Identity correlation
- Role and entitlement cataloging
- Access grant management
- Access request governance
- Access review and certification
- Privileged access governance
- Segregation-of-duties analysis
- Access risk evaluation
- Evidence binding for access decisions
- Identity governance reporting

The engine integrates with all previously completed AQELYN engines while preserving architectural modularity.

39. Major Engineering Decisions

39.1 Decision 1 - Dedicated Identity Governance Engine

Identity governance responsibilities are implemented as a standalone engine rather than embedded directly into the Policy Engine or Compliance Engine.

Rationale:

- Clear separation of identity governance from policy evaluation.
- Independent lifecycle and scaling.
- Improved support for certification and access review campaigns.
- Better support for privileged access and SoD analysis.

39.2 Decision 2 - Identity-to-Account Correlation Model

The engine explicitly separates governed identities from accounts.

Benefits:

- Supports one identity across many systems.
- Supports service accounts and machine identities.
- Enables orphaned account detection.
- Improves traceability of access ownership.

39.3 Decision 3 - Evidence-Backed Access Decisions

Access decisions reference immutable evidence maintained by the AQELYN Evidence Engine.

Benefits:

Access approvals become auditable.
 Certification decisions are evidence-backed.
 Review records remain traceable.
 Compliance requirements can consume identity evidence.

39.4 Decision 4 - Event-Driven Governance

Identity, account, access, certification, and governance state changes are published through the AQELYN Event Bus.

Examples include:

```
identity.created
account.linked
access.requested
access.granted
access.revoked
access.review.completed
certification.started
certification.completed
governance.sod.violation.detected
```

This maintains loose coupling between AQELYN engines.

39.5 Decision 5 - Universal Object Model Extension**New domain objects introduced include:**

```
Identity
Account
Role
Entitlement
AccessGrant
AccessReview
CertificationCampaign
```

These extend the Universal Object Model without modifying existing object definitions.

40. Architectural Integration Summary

| Engine | Integration |

|---|---|

| IS-001 Kernel | Runtime lifecycle and service registration |

| IS-002 Universal Object Model | Identity governance objects |

| IS-003 Event Bus | Identity and access governance events |

| IS-004 Evidence Engine | Approval, review, and certification evidence |

| IS-005 Knowledge Graph | Identity, account, role, entitlement, asset relationships |

| IS-006 Trust Engine | Identity trust, device trust, approval trust |

| IS-007 Mission Engine | Mission-required roles, certifications, clearances |

| IS-008 Workflow Engine | Access request, review, certification, remediation workflows |

| IS-009 Policy Engine | Authorization, approval chain, review frequency, expiration rules |

| IS-010 Compliance Engine | Access control compliance, audit readiness, privileged access review |

No existing engine required redesign.

41. Repository Impact Summary

Repository structure remains unchanged.

Implementation is expected within existing project directories, including:

```
AQELYN/
■■■ src/identity_governance/
■■■ tests/identity_governance/
■■■ api/identity_governance/
■■■ docs/identity_governance/
■■■ archive/
```

No top-level directories were added, removed, or renamed.

42. Security Impact Summary

The specification introduces identity-governance-specific security controls:

```
Policy-driven authorization
Least privilege evaluation
Separation of duties
Privileged access governance
Evidence-backed approval chains
Immutable access review history
Certification campaign governance
Continuous access risk evaluation
```

No reduction in the security posture of existing components was identified.

43. Capabilities Added

The engine enables AQELYN to support:

```
Identity inventory
Account inventory
Identity correlation
Role governance
Entitlement governance
Access grants
Access requests
Access approvals
Access reviews
Access certification
Privileged access governance
Segregation-of-duties detection
Stale access detection
Orphaned account detection
Identity risk reporting
Access compliance reporting
```

44. Risks Identified

| Risk | Mitigation |

|---|---|

| Identity correlation errors | Configurable deterministic matching rules and manual review |

| Orphaned account misclassification | Evidence-backed account ownership and review workflows |

| Privileged access abuse | Privileged access governance, risk scoring, and certification |

| Excessive entitlement accumulation | Periodic review and stale access detection |

| Large certification campaigns | Asynchronous campaign processing and high-volume review handling |

| SoD false positives | Configurable rule sets and escalation workflow |
| Policy evaluation failure | Queue for re-evaluation and require manual review |

No critical architectural risks were identified that require redesign.

45. Verification Summary

The specification defines verification for:

Unit testing
Integration testing
System testing
Security testing
Regression testing

Acceptance criteria cover identity registry, account registry, role and entitlement catalog, access governance workflows, certification lifecycle, privileged access, SoD analysis, risk evaluation, integration with IS-001 through IS-010, repository validation, and testing documentation.

46. Engineering Principles Confirmed

The implementation complies with established AQELYN principles:

Modular architecture
Event-driven communication
Immutable evidence references
Traceability
Explainability
Security by design
Repository stability
Backward compatibility
Governance-before-archive discipline

47. Dependencies

Required:

EA-0001 through EA-0010
IS-001 through IS-010

Enables:

IS-012 and subsequent identity-dependent components

48. Completion Record

Engineering Archive : EA-0011
Implementation Specification : IS-011
Title : AQELYN Identity & Access Governance Engine
Engineering Status : COMPLETE
Repository Status : UNCHANGED
Architecture Status : EXTENDED
Backward Compatibility : MAINTAINED
Engineering Rule :
IS Completed
↓
EA Generated
↓
Continue

49. Archive Index Update

```
EA-0001 IS-001 AQELYN Kernel
EA-0002 IS-002 Universal Object Model
EA-0003 IS-003 AQELYN Event Bus
EA-0004 IS-004 AQELYN Evidence Engine
EA-0005 IS-005 AQELYN Knowledge Graph
EA-0006 IS-006 AQELYN Trust Engine
EA-0007 IS-007 AQELYN Mission Engine
EA-0008 IS-008 AQELYN Workflow Engine
EA-0009 IS-009 AQELYN Policy Engine
EA-0010 IS-010 AQELYN Compliance & Governance Engine
EA-0011 IS-011 AQELYN Identity & Access Governance Engine
```

50. Engineering Phase Status

Completed Engineering Archives : EA-0001 through EA-0011

Current Status:
EA-0011 COMPLETE

Next Implementation Specification:
IS-012

EA-0011 is completed and archived. The engineering workflow is consistent with the project rule:

Implementation Specification -> Engineering Archive -> Continue

From this point onward, the next engineering artifact is IS-012.

51. Engineering Archive Publication Standard

EA-0011 follows the AQELYN Engineering Archive Publication Standard.

Required package structure:

```
EA-xxxx/
■
■ ■ ■ diagrams/
■ ■ ■ ■ Architecture.svg
■ ■ ■ ■ Component.svg
■ ■ ■ ■ Workflow.svg
■ ■ ■ ■ EventFlow.svg
■ ■ ■ ■ Integration.svg
■
■ ■ ■ examples/
■ ■ ■ ■ example_*.md
■
■ ■ ■ HTML/
■ ■ ■ ■ index.html
■ ■ ■ ■ styles.css
■ ■ ■ ■ assets/
■
■ ■ ■ images/
■
■ ■ ■ journal/
■ ■ ■ ■ Engineering_Journal.md
■
■ ■ ■ manifest/
■ ■ ■ ■ manifest.json
■
■ ■ ■ MD/
■ ■ ■ ■ EA-xxxx.md
■
■ ■ ■ PDF/
```

```
■ ■■■ EA-xxxx.pdf
■
■ ■■■ requirements/
■ ■■■ Requirements_Matrix.md
■
■ ■■■ traceability/
■ ■■■ Traceability_Matrix.md
■
■ ■■■ README.md
```

The master Markdown is the source of truth. The PDF and HTML are generated from the same master Markdown and must not omit sections.

52. Requirements Matrix

Requirement ID	Requirement	Evidence in Archive	Status
FR-011-001	Maintain governed identity records	Sections 8, 12	Complete
FR-011-002	Map identities to accounts	Sections 8, 12	Complete
FR-011-003	Maintain role and entitlement catalog	Sections 8, 12, 13	Complete
FR-011-004	Track access grants	Sections 8, 12, 13	Complete
FR-011-005	Support access request workflow	Sections 8, 12, 19	Complete
FR-011-006	Support periodic access reviews	Sections 8, 12, 25	Complete
FR-011-007	Support certification campaigns	Sections 8, 12, 25	Complete
FR-011-008	Govern privileged access	Sections 8, 12, 27	Complete
FR-011-009	Detect orphaned and stale access	Sections 8, 27	Complete
FR-011-010	Identify SoD violations	Sections 8, 12, 27	Complete
NFR-011-001	Traceability	Sections 9, 14, 16, 48	Complete
NFR-011-002	Immutability of approval records	Sections 9, 24, 39	Complete
NFR-011-003	Least privilege	Sections 9, 24, 27	Complete
NFR-011-004	Separation of duties	Sections 9, 12, 24	Complete
NFR-011-005	Auditability	Sections 9, 28, 32	Complete
NFR-011-006	Event-driven synchronization	Sections 9, 15, 30	Complete

53. Traceability Matrix

Source	Target	Relationship
IS-011 Purpose	EA-0011 Objective	Defines why the engine exists
Identity Registry	FR-011-001	Implements governed identity records
Account Registry	FR-011-002	Implements identity-to-account mapping
Role & Entitlement Catalog	FR-011-003	Implements role and entitlement governance
Access Grant Manager	FR-011-004	Tracks access relationships
Access Request Manager	FR-011-005	Implements access request governance
Access Review Engine	FR-011-006	Supports periodic access reviews
Certification Manager	FR-011-007	Implements campaign-based certification
Privileged Access Governance Service	FR-011-008	Governs high-risk access
Access Risk Engine	Risk Evaluation	Calculates access risk
SoD Analyzer	FR-011-010	Detects conflicting privileges

| Evidence Engine Integration | Evidence-backed decisions | Binds access decisions to evidence |

| Policy Engine Integration | Authorization rules | Determines who can request and approve |

| Compliance Integration | IS-010 | Supplies access compliance evidence |

| Event Bus Integration | NFR-011-006 | Publishes identity governance events |

| Repository Validation | Repository Standard | Confirms no top-level redesign |

54. Engineering Journal

Journal Entry - EA-0011

EA-0011 was created to archive completion of IS-011 - AQELYN Identity & Access Governance Engine.

The archive records the expansion of AQELYN into identity governance and access accountability. IS-011 defines the structure needed to manage governed identities, accounts, roles, entitlements, access requests, grants, reviews, certifications, privileged access, segregation-of-duties analysis, risk scoring, and evidence-backed governance decisions.

The engineering design preserves the fixed AQELYN repository structure and maintains backward compatibility with previously completed engines.

Lessons Learned

Identity governance must be modeled separately from policy enforcement. Policies determine authorization and required approval chains, but the Identity & Access Governance Engine owns identity lifecycle, access state, review history, and certification evidence.

Governance Note

EA-0011 follows the master-document publication workflow. The Markdown file is the authoritative source, and PDF/HTML representations are generated from the same content.

55. Examples

55.1 Example Identity

```
identity_id: ID-0001
identity_type: employee
display_name: Jane Doe
owner: HR
status: active
trust_score: 0.92
risk_score: low
```

55.2 Example Account Mapping

```
account_id: ACC-1001
identity_id: ID-0001
platform: Azure AD
username: jane.doe@example.com
status: active
last_seen: 2026-07-07T10:00:00Z
```

55.3 Example Access Grant

```
grant_id: GRANT-2001
identity: ID-0001
```

```
account: ACC-1001
role: ROLE-FINANCE-APPROVER
entitlement: ENT-FINANCE-PAYMENT-APPROVE
approval: GOV-DEC-3001
expires_at: 2026-12-31
```

55.4 Example SoD Violation Event

```
{
  "event_type": "governance.sod.violation.detected",
  "identity_id": "ID-0001",
  "conflict": ["create_vendor", "approve_vendor_payment"],
  "severity": "high",
  "source_engine": "aqelyn_identity_access_governance_engine"
}
```

56. Manifest Summary

Archive contents include:

```
README.md
MD/EA-0011.md
PDF/EA-0011.pdf
HTML/index.html
HTML/styles.css
manifest/manifest.json
requirements/Requirements_Matrix.md
traceability/Traceability_Matrix.md
journal/Engineering_Journal.md
diagrams/Architecture.svg
diagrams/Component.svg
diagrams/Workflow.svg
diagrams/EventFlow.svg
diagrams/Integration.svg
examples/example_identity_governance.md
```

57. Final Archive Statement

EA-0011 is the Engineering Archive for IS-011 - AQELYN Identity & Access Governance Engine.

It records the completed specification, the architectural decisions, the integration model, the repository impact, the risk posture, verification requirements, acceptance criteria, archive index update, and the engineering publication standard.

```
EA-0011 COMPLETE
IS-011 COMPLETE
NEXT: IS-012
```